

LINA ANAYA

Data Scientist

Tarragona, Spain | LinkedIn | Google Scholar | Website

Profile Summary

Data scientist with two years of applied machine learning experience building production-track anomaly detection, threat classification, and LLM-orchestrated automation pipelines for a cybersecurity startup, on a foundation of a PhD heavily focused on applied econometrics and causal inference, alongside six years of quantitative research. Built and deployed cloud-based ML pipelines on Google Vertex AI reaching 98–99% model accuracy, designed a statistical model-validation method reused across six production pipelines, and built an LLM pipeline that cut regulation-to-rule turnaround from months to days. Comfortable owning a problem end-to-end — framing it, building it, and presenting results to technical and non-technical stakeholders alike.

Skills

Programming & Data: Python, SQL, Polars, Pandas, NumPy

Machine Learning: Scikit-learn, YDF (Gradient Boosted Trees, Isolation Forest), TensorFlow/Keras, LoRA fine-tuning (Unsloth, PEFT, TRL), hyperparameter optimisation

Statistics & Causal Inference: Hypothesis testing, bootstrap confidence intervals, Monte Carlo simulation, difference-in-differences, RCTs/quasi-experimental design, econometrics

LLM & GenAI: LiteLLM, Instructor, retrieval-augmented generation, prompt engineering, GGUF quantisation (llama.cpp)

Cloud & MLOps: Google Cloud Platform (Vertex AI, BigQuery, Cloud Storage), Kubeflow Pipelines, Docker, GitHub Actions CI/CD, Prefect

Software Engineering: FastAPI, Pydantic, SQLAlchemy/Alembic, pytest, Git

Languages: Spanish (native), English (C2)

Work Experience

Periphery Security (formerly Qomodo) — United Kingdom — remote

Data Scientist | Mar 2024 – Jun 2026

- Built and validated six machine learning pipelines for detecting anomalous and malicious device network/process activity, solving the lack of labelled attack data with synthetic injection, public-dataset fusion, and honeypot-traffic techniques; reached accuracy in the high 90s with a false-positive rate below 1% on internal test data.
- Designed a statistical model-deployment gate, hypothesis testing combined with bootstrap confidence intervals, reused across every detection pipeline and deployed to Google Cloud Vertex AI endpoints.
- Built an internal LLM-orchestrated pipeline (Prefect, LiteLLM/Instructor, FastAPI) that extracts compliance controls from regulations and drafts matching audit rules, cutting turnaround from months to days; diagnosed and fixed a production bug in its review workflow.
- Designed a device cybersecurity risk-scoring engine spanning three regulatory frameworks (NIST SP 800-53, NIST SP 800-171, ISA/IEC 62443), with a CVE risk formula validated by a five-million-row Monte Carlo simulation.
- Fine-tuned and quantised small language models (LoRA, GGUF) into an offline security-log triage assistant, evaluated with a custom ROUGE/BLEU/METEOR/BERTScore harness.
- Coordinated deployment formats with the cloud and embedded engineering team as the company's sole data scientist, presenting results company-wide, including to the CEO, CFO, and sales team.

Ekology DataConsulting — Spain

Founder | Jan 2023 – Mar 2024

- Founded a data-literacy consultancy for small business owners, delivering personalised training and building an Instagram channel to share data-driven decision-making content; partnered with Escuela de Emprendimiento E3 and the Colombian Consulate to deliver a free data-decision-making course reaching over 100 entrepreneurs across Catalonia and Andorra.

University of Leeds and University of Bradford — United Kingdom — remote

Postdoctoral Research Associate, Data Scientist | May 2021 – Sept 2022

- Designed a difference-in-differences quasi-experimental model estimating the causal impact of the UK's first COVID-19 lockdown on mental health, finding effects equivalent to 60% of the impact of unemployment, with a 50% decline for people in overcrowded housing and disproportionate effects on women and ethnic minorities.

University of Arkansas — USA — remote/hybrid

Data Science Graduate Research Assistant | Aug 2017 – May 2021

- Used linear regression and panel-data models on PISA and university datasets to quantify gender gaps in maths/STEM outcomes, finding highly skilled women 21–24% less likely to major in STEM and effort differences explaining up to 40% of the science achievement gap.
- Analysed PISA 2009–2015 data with a random-effects model to show that difficulty-first question ordering increased exam dropout and reduced correct answers, informing exam-design recommendations.
- Taught as a Teaching Assistant in Statistics, Calculus, and Econometrics (Aug 2018 – May 2021), supporting college and PhD-level students with questions and problem-set feedback.

Education

University of Arkansas — Fayetteville, USA

PhD, Education Economics (Education Policy), May 2021

4Geeks Academy — Spain

Data Science & Machine Learning, Feb 2024

ICESI University — Cali, Colombia

BA, Economics and International Business, Aug 2015

Published Research Articles

- Anaya, L., Zamarro, G. (2023). The role of student effort on performance in PISA: Revisiting the gender gap in achievement. [Oxford Economic Papers](#).
- Anaya, L., Howley, P., Waqas, M., Yalonetzky, G. (2023). Locked down in distress: a causal estimation of the mental-health fallout from the COVID-19 pandemic in the UK. [Economic Inquiry](#).
- Anaya, L., Iriberry, N., Rey-Biel, P., Zamarro, G. (2022). Understanding performance in test taking: The role of question difficulty order. [Economics of Education Review](#).
- Anaya, L., Stafford, F., Zamarro, G. (2021). Gender gaps in math performance, perceived mathematical ability and college STEM education: the role of parental occupation. [Education Economics](#).

Conference Presentations & Talks

8 selected/invited presentations, 2018–2023, including AERA (Canada), AEPF (USA × 3), the 44th SAEe Symposium (Spain), EERA Emerging Researchers (Switzerland), Escuela de Emprendimiento E3 & the Colombian Consulate (Spain), and the Arkansas Environmental Education Association (USA).